Importing Required Packages

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```
!pip install mido
```

```
Requirement already satisfied: mido in /usr/local/lib/python3.10/dist-packages (1.3.3)
Requirement already satisfied: packaging in /usr/local/lib/python3.10/dist-packages (from mido) (24.2)
```

```
import mido
from mido import MidiFile, MidiTrack, Message
from keras.layers import LSTM, Dense, Activation, Dropout, Flatten
from keras.preprocessing import sequence
from keras.models import Sequential
from tensorflow.keras.optimizers import Adam
from sklearn.preprocessing import MinMaxScaler, StandardScaler
import numpy as np
```

Load MID file/files

```
#download zip dari vclass, upload hasil download ke /content/
```

```
!unzip /content/chillhopdata.zip -d chillhop
```

```
unzip:  cannot find or open /content/chillhopdata.zip, /content/chillhopdata.zip.zip or /content/chillhopdata.zip.ZIP.
```

```
import os

notes = []
for song in os.listdir("/content/chillhop"):
  mid = MidiFile('/content/chillhop/' + song)
  for msg in mid:
    if not msg.is_meta and msg.channel == 0 and msg.type =='note_on':
      data = msg.bytes()
      notes.append(data[1])
```

Scale Data

```
scaler = MinMaxScaler()
notes = list(scaler.fit_transform(np.array(notes).reshape(-1, 1)))
```

Create Train Data

```
notes = [list(note) for note in notes]

X = []
y = []

n_prev=30

for i in range(len(notes) - n_prev):
  X.append(notes[i:i+n_prev])
  y.append(notes[i+n_prev])
```

```
X_test = X[-300:]
X = X[:-300]
y = y[:-300]
```

Build LSTM

```
model = Sequential()
model.add(LSTM(256, input_shape=(n_prev, 1), return_sequences=True))
model.add(Dropout(0.6))
model.add(LSTM(128, input_shape=(n_prev, 1), return_sequences=True))
model.add(Dropout(0.6))
model.add(LSTM(64, input_shape=(n_prev, 1), return_sequences=False))
model.add(Dropout(0.6))
model.add(Dense(1))
model.add(Activation('linear'))
model.summary()

optimizer = Adam(learning_rate=0.001)
model.compile(loss='mse', optimizer=optimizer)
```

```
/usr/local/lib/python3.10/dist-packages/keras/src/layers/rnn/rnn.py:204: UserWarning: Do not pass an `input_shape`/`input_dim` argument
  super().__init__(**kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 30, 256)	264,192
dropout (Dropout)	(None, 30, 256)	0
lstm_1 (LSTM)	(None, 30, 128)	197,120
dropout_1 (Dropout)	(None, 30, 128)	0
lstm_2 (LSTM)	(None, 64)	49,408
dropout_2 (Dropout)	(None, 64)	0
dense (Dense)	(None, 1)	65
activation (Activation)	(None, 1)	0

Total params: 510,785 (1.95 MB)
Trainable params: 510,785 (1.95 MB)

Training

```
model.fit(np.array(X), np.array(y), batch_size=16, epochs=10, verbose=1)
```

```
Epoch 1/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 31s 134ms/step - loss: 0.0303
Epoch 2/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 41s 133ms/step - loss: 0.0192
Epoch 3/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 41s 134ms/step - loss: 0.0168
Epoch 4/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 26s 135ms/step - loss: 0.0168
Epoch 5/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 42s 141ms/step - loss: 0.0161
Epoch 6/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 39s 134ms/step - loss: 0.0144
Epoch 7/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 41s 132ms/step - loss: 0.0150
Epoch 8/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 41s 133ms/step - loss: 0.0147
Epoch 9/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 26s 134ms/step - loss: 0.0142
Epoch 10/10
195/195 ━━━━━━━━━━━━━━━━━━━━ 25s 130ms/step - loss: 0.0140
<keras.src.callbacks.history.History at 0x7cabccf58c40>
```

Generating & Saving LSTM Music

```
prediction = model.predict(np.array(X_test))
prediction = np.squeeze(prediction)
prediction = np.squeeze(scaler.inverse_transform(prediction.reshape(-1,1)))
prediction = [int(i) for i in prediction]

mid = MidiFile()
track = MidiTrack()
t = 0
for note in prediction:
    # 147 means note_on
    # 67 is velosity
    note = np.asarray([147, note, 67])
    bytes = note.astype(int)
    msg = Message.from_bytes(bytes[0:3])
    t += 1
    msg.time = t
    track.append(msg)
mid.tracks.append(track)
mid.save('LSTM_music.mid')
```

```
10/10 ━━━━━━━━━━━━━━━━━━━━ 3s 237ms/step
```

Start coding or generate with AI.